

Activity 11 Matrix arithmetic

1.

a) $\begin{bmatrix} 4 & 6 \\ 1 & 5 \end{bmatrix}$

b) $\begin{bmatrix} 9 & 3 \\ -3 & 6 \end{bmatrix}$

c) $\begin{bmatrix} 2 & 10 \\ 4 & 6 \end{bmatrix}$

d) $\begin{bmatrix} 11 & 13 \\ 1 & 12 \end{bmatrix}$

e) Not possible *invalid dimension*
The matrices have different sizes, i.e. C has 3 rows and A has 2 rows.

f) Not possible *invalid dimension* The matrices have different sizes.

g) $\begin{bmatrix} 5 & 18 \\ 3 & 1 \end{bmatrix}$

h) $\begin{bmatrix} -2 & 11 \\ 3 & 8 \end{bmatrix}$

i) BC seen as a single variable. $B \times C$ is not possible.

j) $\begin{bmatrix} -7 & -3 \\ -5 & 7 \\ 5 & 13 \end{bmatrix}$

k) Not possible *invalid dimension*. The number of columns in the first matrix is not the same as the number of rows in the second matrix.

l) $\begin{bmatrix} 8 & 5 \\ -5 & 3 \end{bmatrix}$

m) Not possible *invalid dimension* $B \times A$ is a 2×2 matrix whereas C is 3×2 .

